

GANG HE

Office: 135 East 22nd Street, Room 615, New York, NY 10010
Email: gang.he@baruch.cuny.edu; Phone: +1 (646) 660-6716
Web: drganghe.github.io; Lab: deeppolicylab.github.io

RESEARCH INTEREST

Energy systems, energy economics, energy and climate policy, energy transition

CURRENT APPOINTMENT

Baruch College, City University of New York	Marxe School of Public and International Affairs
Associate Professor	Aug. 2025 – present
Assistant Professor	Aug. 2023 – Aug. 2025

AFFILIATION

Earth and Environmental Sciences, CUNY Graduate Center	Jan. 2025 – present
CUNY Institute for Demographic Research	Aug. 2023 – present
Department of Technology, AI, and Society, Stony Brook University	Aug. 2023 – present
Energy Technology Area, Lawrence Berkeley National Laboratory	Aug. 2015 – present

PREVIOUS APPOINTMENTS

Stony Brook University	Department of Technology and Society
Assistant Professor	Aug. 2015 – Aug. 2023
University of California, Berkeley	Renewable and Appropriate Energy Laboratory
Lawrence Berkeley National Laboratory	Energy Technology Area
Graduate Student Researcher	Aug. 2012 – Aug. 2015
Stanford University	Program on Energy and Sustainable Development
Research Associate	Aug. 2008 – Aug. 2010

EDUCATION

University of California, Berkeley	2015
Ph.D. in Energy and Resources	Advisor: Prof. Daniel Kammen
Dissertation: <i>Decarbonizing China's Power Sector: Potential, Prospect, and Policy</i>	
Columbia University	2008
M.A. in Climate and Society	
Peking University	2006
B.S. and M.S. in Geography	

CAREER HIGHLIGHTS

- **Research Visibility:** Author of 60+ peer-reviewed publications in leading interdisciplinary and field-defining journals, including *Nature*, *Nature Communications*, *Nature Energy*, *Nature Water*, *One Earth*, *Environmental Science & Technology*, *Energy Policy*, *Renewable and Sustainable Energy Reviews*. Research featured by major science and policy media, including *Nature*, *The New York Times*, *The Economist*, *Scientific American*, *Carbon Brief*, *National Geographic*, *E&E News*.
- **Google Scholar Citations:** 7500+, h-index: 35, i-10 index: 65, reflecting sustained and growing influence across energy systems modeling, climate policy, and clean energy transitions.
- **Grants:** PI on US\$1.5M+ in competitively awarded research funding from federal (NSF, SSA, LBNL), state (NYDEC), philanthropic (Sloan, ClimateWorks, Sequoia, Growald), NGO (EDF), and industrial sources, supporting interdisciplinary research on decarbonization, energy and climate policy, and clean energy transition.
- **Thought Leadership:** Delivered 20+ invited talks at major universities, government agencies, and international institutions, including the U.S. Energy Information Administration, Asia Pacific Energy Research Centre, and International Association of Energy Economics.
- **Policy Engagement and Global Impact:** Testified for the New York State Climate Leadership and Community Protection Act (CLCPA); advised the New York State Climate Action Council's Scoping Plan; and served on the U.S.-China Working Group on Enhancing Climate Action in the 2020s (Energy Transition Sub-Working Group). Research has attracted over 90 [policy citations and mentions](#) by 54 leading policy institutions including IPCC(AR6), IIASA, World Bank, UNEP, IRENA, WTO, SEI, IISD, CSIS, and Brookings.

DISTINCTIONS

CUNY Academy Sandi Cooper Award for Outstanding Research for Associate Professors	2026
National Committee on United States-China Relations Public Intellectuals Program Fellow	2025
Stanford/Elsevier Top 2% Scientists List 2025, 2024	2025
Baruch College Faculty Research Award	2024
Integrated Assessment Modeling Consortium (IAMC) Best Poster Award	2022
ITIF Energy Innovation Policy and Management Scholar	2019
Institute for New Economic Thinking Young Scholar	2013
Aspen Environment Forum Scholar	2011
World Resources Institute Cynthia Helms Fellow	2008
Asia Society Asia 21 Young Leaders	2007

PUBLICATIONS

* denotes corresponding author, __ denotes student advisee, ~ denotes visiting scholar

Complete list of publications is available at:

 orcid.org/0000-0002-8416-1965  [Google Scholar](#)  [Research Gate](#)

SELECTED WORK

13. **He, Gang*** and Jiang Lin. 2026. [How Hard Are Hard-to-Abate Sectors? Rethinking Industrial Decarbonization Pathways](#). *One Earth* 9(8): 101784.

12. **He, Gang***. 2026. [Renewable integration and AI demand reshaped power grids in 2025](#). *Nature Reviews Clean Technology* 2(1): 13-14.
11. Qiu, Minghao, **Gang He***, and Peter Marcotullio. 2025. [Imported Solar Photovoltaics Contributed to Health and Climate Benefits in the United States](#). *One Earth* 8 (11): 101467.
Coverage: [CUNY News](#), [PV Magazine](#)
10. Liu, Shangwei*, **Gang He***, Minghao Qiu, and Daniel M. Kammen. 2025. [Can China break the 'cost curse' of nuclear power?](#). *Nature* 643: 1186–88.
Coverage: [Harvard News](#), [CUNY News](#), [JHU News](#), [The Economist](#), [The New York Times](#)
9. Helveston, John, **Gang He***, Michael Davidson. 2022. [Quantifying the cost savings of global photovoltaic supply chains](#). *Nature*. 612: 83–87.
Award: [IAMC 2022 Best Poster Award](#)
Coverage: [SBU News](#), [GWU News](#), [UCSD News](#), [Nature Research Highlights](#), [E&E News](#), [Foreign Policy](#), [PV Magazine](#), [dot.LA](#), [pvbuzz](#), [Innovate Long Island](#), [Mercom India](#), [China News](#)
Invited presentation: [Peking University](#), [Tsinghua University](#), [Harvard University](#) (M.D.), [ITIF](#) (J.H.), [Wuhan University](#), [Climate Change Economics Forum](#)
8. **He, Gang***, Jiang Lin*, Froylan Sifuentes, Xu Liu, Nikit Abhyankar, and Amol Phadke*. 2020. [Rapid Cost Decrease of Renewables and Storage Accelerates the Decarbonization of China's Power System](#). *Nature Communications*. 11(1):2486.
Coverage: [Nature](#), [CarbonBrief](#) [Headline](#), [Inside Climate News](#), [Forbes](#), [SBU News](#), [LBL News](#)
Invited presentation: [Asia Pacific Energy Research Centre \(APEREC\) Annual Conference 2020](#), [Beijing Institute of Technology](#), [Dartmouth College](#), [Harvard Kennedy School Growth Lab](#).
7. **He, Gang***, Jiang Lin, Ying Zhang, Wenhua Zhang, Guilherme Larangeira, Chao Zhang, Wei Peng, Manzhi Liu, and Fuqiang Yang. 2020. [Enabling a Rapid and Just Transition Away from Coal in China](#). *One Earth*. 3(2):187–194.
Coverage: [Nature](#), [SBU News](#)
Invited presentation: [Coal Transition Network](#), [Peking University](#)
6. **He, Gang***, David Victor. 2017. [Experiences and lessons from China's success in providing electricity for all](#). *Resources, Conservation and Recycling*. 122:335–338.
Invited presentation: [2017 EIA Energy Conference](#), [2017 IAEE International Conference](#)
5. **He, Gang***, Anne-Perrine Avrin, James H. Nelson, Josiah Johnston, Ana Mileva, Jianwei Tian, and Daniel M. Kammen*. 2016. [SWITCH-China: A Systems Approach to Decarbonizing China's Power System](#). *Environmental Science & Technology*. 50(11):5467–5473.
Invited presentation: [Carnegie Mellon University](#), [Lawrence Berkeley National Laboratory](#), [Michigan State University](#).
4. **He, Gang***, and Daniel M. Kammen*. 2016. [Where, When and How Much Solar Is Available? A Provincial-Scale Solar Resource Assessment for China](#). *Renewable Energy*. 85:74–82.
3. Zhou, Nan*, **Gang He**, Christopher Williams, and David Fridley. 2015. [ELITE Cities: A Low-Carbon Eco-City Evaluation Tool for China](#). *Ecological Indicators*. 48:448–56.
Coverage: [The Guardian](#)
2. **He, Gang***, and Daniel M. Kammen. 2014. [Where, When and How Much Wind Is Available? A](#)

Provincial-Scale Wind Resource Assessment for China. *Energy Policy*. 74:116–122.

1. He, Gang*, and Richard Morse. 2013. Addressing Carbon Offsetters' Paradox: Lessons from Chinese Wind CDM. *Energy Policy*. 63:1051–1055.

Coverage: *Caixin*

Invited presentation: UNFCCC CDM Executive Board meeting

SELECTED JOURNAL PAPER BY TOPIC

Clean Energy Supply Chain

3. Qiu, Minghao, Gang He*, and Peter Marcotullio. 2025. Imported Solar Photovoltaics Contributed to Health and Climate Benefits in the United States. *One Earth* 8 (11): 101467.
2. Helveston, John, Gang He*, Michael Davidson. 2022. Quantifying the cost savings of global photovoltaic supply chains. *Nature*. 612: 83–87.
1. Chen, Shi, Xi Lu*, Chris P. Nielsen, Michael B. McElroy*, Gang He, Shaohui Zhang, Kebin He, Xiu Yang, Fang Zhang, and Jimin Hao. 2023. Deploying Solar Photovoltaic Energy First in Carbon-Intensive Regions Brings Gigatons More Carbon Mitigations to 2060. *Communications Earth & Environment* 4: 369.

Power System Modeling and Clean Power Transition

11. Liu, Junhong, Fredrich Kahrl, Liqun Peng, Jiang Lin*, and Gang He. 2026. Toward Efficient Resource Adequacy: The Role of Shared Reserves in China's Power Grid. *Energy Strategy Reviews* 63: 102043.
10. Lu, Zhenwei, Bin Ye*, Zhe Yu, Gang He, and Shuai Shao. 2026. Emission Trading Scheme Reshapes the Decarbonisation Pathways of China's Power Sector. *Applied Energy* 412: 127726.
9. Peng, Liqun, Yang Guo, Shangwei Liu, Gang He, and Denise L. Mauzerall*. 2024. Subsidizing Grid-Based Electrolytic Hydrogen Will Increase Greenhouse Gas Emissions in Coal Dominated Power Systems. *Environmental Science & Technology* 58 (12): 5187–5195.
8. Peng, Liqun, Denise L. Mauzerall*, Yaofeng D. Zhong, and Gang He*. 2023. Heterogeneous Effects of Battery Storage Deployment Strategies on Decarbonization of Provincial Power Systems in China. *Nature Communications* 14 (1): 4858.
7. Lin, Jiang*, Nikit Abhyankar, Gang He, Xu Liu, and Shengfei Yin. 2022. Large Balancing Areas and Dispersed Renewable Investment Enhances Grid Flexibility in a Renewable-Dominant Power System. *iScience*. 25(2):103749.
6. Li, Bo, Ziming Ma, Patricia Hidalgo-Gonzalez, Alex Lathem, Natalie Fedorova, Gang He, Haiwang Zhong, Minyou Chen*, and Daniel M. Kammen*. 2021. Modeling the Impact of EVs in the Chinese Power System: Pathways for Implementing Emissions Reduction Commitments in the Power and Transportation Sectors. *Energy Policy*. 149:111962.
5. Li, Haoran, Xueqin Cui, Jingxuan Hui, Gang He, Yuwei Weng, Yaoyu Nie, Can Wang, and Wenjia Cai*. 2021. Catchment-Level Water Stress Risk of Coal Power Transition in China under 2°C/1.5°C Targets. *Applied Energy*. 294: 116986.
4. Li, Bo, Minyou Chen*, Ziming Ma, Gang He, Wei Dai, Dongran Liu, Chi Zhang, and Haiwang Zhong. 2021. Modelling Integrated Power and Transportation Sectors Decarbonization with Hydrogen Energy Storage. *IEEE Transactions on Industry Applications*, 58(2):2677–2693.
3. He, Gang*, Jiang Lin*, Froylan Sifuentes, Xu Liu, Nikit Abhyankar, and Amol Phadke*. 2020. Rapid Cost Decrease of Renewables and Storage Accelerates the Decarbonization of China's Power System. *Nature Communications*. 11(1):2486.

2. **He, Gang***, Hongliang Zhang*, Yuan Xu*, and Xi Lu*. 2017. [China's Clean Power Transition: Current Status and Future Prospect](#). *Resources, Conservation and Recycling*, Environmental Challenges and Potential Solutions of China's Power Sector, 121:3–10.
1. **He, Gang***, Anne-Perrine Avrin, James H. Nelson, Josiah Johnston, Ana Mileva, Jianwei Tian, and Daniel M. Kammen*. 2016. [SWITCH-China: A Systems Approach to Decarbonizing China's Power System](#). *Environmental Science & Technology*. 50(11):5467–5473.

Climate Change and Energy Systems Resilience

2. Wang, Jiarui, Junqi Liu, Lei Zhu*, Gang He*. 2026. [Resilience planning for power systems under deep climate uncertainty](#). *Energy Economics* 161: 109527.
1. Liu, Laibao*, **Gang He**, Mengxi Wu*, Gang Liu, Haoran Zhang, Ying Chen, Jiashu Shen, Shuangcheng Li. 2023. [Climate change impacts on planned supply–demand match in global wind and solar energy systems](#). *Nature Energy* 8(8): 870–80.

Energy and Water Nexus

3. Yue Qin*, Yaoping Wang, Shiyu Li, Hang Deng, Niko Wanders, Joyce Bosmans, Liangdian Huang, Chaopeng Hong*, Edward Byers, Daniel Gingerich, Jeff M. Bielicki, **Gang He**. 2023. [Global assessment of the carbon–water tradeoff of dry cooling for thermal power generation](#). *Nature Water* 1(8): 682–693.
2. Zhang, Chao, **Gang He***, Josiah Johnston, and Lijin Zhong. 2021. [Long-Term Transition of China's Power Sector under Carbon Neutrality Target and Water Withdrawal Constraint](#). *Journal of Cleaner Production* 329:129765.
1. Avrin, Anne-Perrine, **Gang He**, and Daniel M. Kammen. 2015. [Assessing the Impacts of Nuclear Desalination and Geoengineering to Address China's Water Shortages](#). *Desalination*. 360:1–7.

Renewable Resources Assessment

4. Liu, Laibao*, Yang Wang*, Zheng Wang, Shuangcheng Li*, Jiangtao Li, **Gang He**, Yan Li, et al. 2022. [Potential Contributions of Wind and Solar Power to China's Carbon Neutrality](#). *Resources, Conservation and Recycling*. 180:106155.
3. Liu, Laibao, Zheng Wang*, Yang Wang, Jun Wang, Rui Chang, **Gang He**, Wenjun Tang, Ziqi Gao, Jiangtao Li, Changyi Liu, Lin Zhao, Dahe Qin, Shuangcheng Li*. 2020. [Optimizing Wind/Solar Combinations at Finer Scales to Mitigate Renewable Energy Variability in China](#). *Renewable and Sustainable Energy Reviews*. 132:110151.
2. **He, Gang***, and Daniel M. Kammen*. 2016. [Where, When and How Much Solar Is Available? A Provincial-Scale Solar Resource Assessment for China](#). *Renewable Energy*. 85:74–82.
1. **He, Gang***, and Daniel M. Kammen. 2014. [Where, When and How Much Wind Is Available? A Provincial-Scale Wind Resource Assessment for China](#). *Energy Policy*. 74:116–122.

Low Carbon Development and Climate Policy

7. Liu, Shangwei*, **Gang He***, Minghao Qiu, and Daniel M. Kammen. 2025. [Can China break the 'cost curse' of nuclear power?](#). *Nature* 643: 1186–88.
6. Liu, Zhu*, Zhu Deng, **Gang He**, Hailin Wang, Xian Zhang, Jiang Lin, Ye Qi, and Xi Liang. 2022. [Challenges and Opportunities for Carbon Neutrality in China](#). *Nature Reviews Earth & Environment*. 3(2):141–155.
5. Lee, Mekyung*, and **Gang He**. 2021. [An Empirical Analysis of Applications of Artificial Intelligence Algorithms in Wind Power Technology Innovation during 1980-2017](#). *Journal of Cleaner Production*.

297:126536.

4. Xu, Jia, Xiujie Tan*, **Gang He**, and Yu Liu. 2019. [Disentangling the Drivers of Carbon Prices in China's ETS Pilots — An EEMD Approach](#). *Technological Forecasting and Social Change*. 139:1–9.
3. Bai, Hongtao, Xiangyu Feng, Huimin Hou, **Gang He***, Yan Dong, and He Xu*. 2018. [Mapping Inter-Industrial CO2 Flows within China](#). *Renewable and Sustainable Energy Reviews*. 93:400–408.
2. Zhou, Nan*, **Gang He**, Christopher Williams, and David Fridley. 2015. [ELITE Cities: A Low-Carbon Eco-City Evaluation Tool for China](#). *Ecological Indicators*. 48:448–56.
1. **He, Gang***, and Richard Morse. 2013. [Addressing Carbon Offsetters' Paradox: Lessons from Chinese Wind CDM](#). *Energy Policy*. 63:1051–1055.

Coal and Just Transition

5. Danneaux, Augustin*, Thomas Le Gallic, Julien Lefèvre, Saritha Sudharmma Vishwanathan, and **Gang He**. 2026. [Coal Mine Workers in the Climate Transition: Subnational Vulnerability Disparities in China and India](#). *Environmental Research Letters* 21(3): 34026.
4. Wu, Huihuang, Junfeng Liu, Xiurong Hu, **He, Gang**, Yuhang Zhou, Xian Wang, Ying Liu, Wenhua Zhang, Jianmin Ma, and Shu Tao. 2024. [Fewer than 15% of coal power plant workers can easily shift to green job by 2060](#). *One Earth*. 7(11): 1994–2007.
3. **He, Gang***, Jiang Lin, Ying Zhang, Wenhua Zhang, Guilherme Larangeira, Chao Zhang, Wei Peng, Manzhi Liu, and Fuqiang Yang. 2020. [Enabling a Rapid and Just Transition Away from Coal in China](#). *One Earth*. 3(2):187–194.
2. Liu, Manzhi*, Meng Chen, and **Gang He***. 2017. [The Origin and Prospect of Billion-Ton Coal Production Capacity in China](#). *Resources, Conservation and Recycling*. 125:70–85.
1. **He, Gang***, Richard Morse. [China's Coal Import Behavior and Its Impacts to Global Energy Market](#), in: Globalization, Development and Security in Asia Volume 3: *The Political Economy of Energy*. Singapore: World Scientific Publishing. 2014. 69–85.

GRANTS

16. “Made in America? Unpacking the Drivers and Impacts of Domestic Clean Energy Manufacturing”. Sponsor: Alfred P. Sloan Foundation. PI: Gang He. Co-PIs: Michael Davidson, Ahmad Lashkaripour, Kaifang Luo, Ilaria Mazzocco, Minghao Qiu. Period: 12/2025–11/2028. Funding: \$750,000.
15. “Clean energy projects and housing value in New York State”. Sponsor: Baruch College Office of Provost Interdisciplinary Innovative Seed Grant. PI: Gang He. Co-PI: Maggie Hu. Period: 12/2025–6/2026. Funding: \$5,000.
14. “Global Clean Energy Supply Chains Analysis and Power Systems Resources Adequacy Assessment”. Sponsor: University of California, Berkeley. CUNY PI: Gang He. Period: 11/2024–10/2026. Funding: \$100,000.
13. “Exploring Energy Burden Among the Older Adults and People with Disabilities in New York City Public Housing Communities”. Sponsor: Social Security Administration. PI: Gang He. Co-PIs: Kaifang Luo, Hilary Botein, Frank Heiland. Period: 10/2024–9/2025. Funding: \$100,000.
12. “Power system beyond coal”. Sponsor: Growald Climate Fund. CUNY PI: Gang He. Period: 8/2024–7/2025. Funding: \$64,000.
11. “Modeling and Training for the Multi-Country Electricity Transition Potential and Challenges Project”, Sponsor: Environmental Defense Fund, CUNY PI: Gang He, Period: 10/2023–9/2024, Funding: \$40,000.

10. “The carbon mitigation, air quality, and human health benefits achieved by global solar PV supply chains”. Sponsor: ClimateWorks Foundation. PI: Gang He. Period: 7/2023–6/2025. Funding: \$100,000.
9. “Analysis of the role of offshore wind and pumped-hydro storage in future power system in China”, Sponsor: Lawrence Berkeley National Laboratory, CUNY PI: Gang He, Period: 5/2023–8/2024, Funding: \$50,000.
8. “Modeling training for the open decarbonization pilot project”, Sponsor: Environmental Defense Fund, PI: Gang He, Period: 5/2022–8/2023, Funding: \$15,000.
7. “Power system scenario model testing and analysis”, Sponsor: Lawrence Berkeley National Laboratory, PI: Gang He, Period: 07/2020–12/2021, Funding: \$50,000.
6. “New York State Solid Waste Characterization”, Sponsor: New York State Department of Environmental Conservation, PI: David Tonjes, Co-PIs: Elizabeth Hewitt, Gang He, Period: 8/2019–8/2024, Funding: \$4,235,776. Share: \$200,000.
5. “Workshop on Data Science Across the Undergraduate Curriculum: University-Industry Online Case Studies on Applications of Data Science”, Sponsor: National Science Foundation, PI: Gang He (David Ferguson), Co-PIs: Thomas Woodson, Elizabeth Hewitt, Marianna Savoca, Period: 10/2019–9/2020, Funding: \$49,796.
4. “Jobs and economic impact of offshore wind development”, Sponsor: Vineyard Wind LLC, PI: Gang He, Period: 12/2018–12/2019, Amount: \$30,000.
3. “Modeling power systems with aggressive renewable costs decline”, Sponsor: Lawrence Berkeley National Laboratory, PI: Gang He, Period: 07/2018–06/2019, Funding: \$40,000.
2. “China’s super efficient appliances policy review”, Sponsor: Lawrence Berkeley National Laboratory, PI: Gang He, Period: 07/2013–12/2013, Funding: \$5,000.
1. “Clean power pathway to 2030 in China”, Sponsor: Lawrence Berkeley National Laboratory, PI: Gang He, Period: 07/2012–12/2012, Funding: \$5,000.

TEACHING

Baruch College, City University of New York:

PAF 3195 [Energy, Climate, and Society](#) (Fall 2023; Spring 2025/2026)

PAF 9174 [Program Evaluation](#) (Spring 2024/2025; Fall 2026)

PAF 9187 [Energy and Climate Policy](#) (Fall 2023/2024/2025)

Stony Brook University:

EST 625 Advanced Science, Technology, Innovation Systems, and Policy (Spring 2022/2023)

EST 607 Energy and Environmental Markets (Spring 2016)

EST 603 Energy Systems Analysis (Spring 2017/2021; [Fall 2022](#))

EST 601 Grand Challenges in Energy and Environmental Policy (Fall 2015/2016/2017/2022)

EST 592 Sustainable Energy (Fall 2018/2019)

EST 535 Electric Power Systems (Spring 2018; Fall 2021)

EST 441 Interdisciplinary Senior Project (Fall 2021)

EST 440 Interdisciplinary Research Methods (Fall 2018/2019/2020; Spring 2022/2023)

EST 393 Project Management (Spring 2018/2019/2020/2021)

Teaching Assistant:

University of California, Berkeley, ER 100/200 Energy and Society (Fall 2011/2013)

Guest Lectures:

Stanford University, CEE 276F China Energy Systems (Jan 2010/Feb 2012/Feb 2014)

University of Copenhagen, LNAK 10069 Climate Change Impacts, Adaptation and Mitigation (Spring 2010/2012/2014)

ADVISING

Postdoctoral Researcher:

Kaifang Luo (PhD in Public Policy from the School of Public Policy, University of Maryland)

PhD Students:

Guilherme Larangeira (GEM Fellowship; Turner Fellowship; ARPA-E 2022 Summer Intern)

Nabil Haque (graduated in 2020; co-advising with Dr. Gerald Stokes; Dissertation: Approaching Sustainability Transition through Climate Technology Projects: Theoretical Underpinning to Understand Past Efforts and Future Outlook)

Mekyung Lee (graduated in 2019; Dissertation: A Study on the Effects of Artificial Intelligence (AI) on Energy Technology Innovation and on Evaluation of Innovation Policy: An Empirical Analysis of Wind Power Technology and Electric Vehicle Technology)

Raphael Apeaning (graduated in 2019; Co-advising with Dr. Gerald Stokes; Dissertation: Technological and Socio-economic Feasibility of Climate Mitigation: a Focus on Developing Economies)

PhD Dissertation Committee:

Aaryan Manoj Nair (Role: Committee Member, CUNY Graduate Center)

Maeen Md. Khairul Akter (Role: Committee Member, Stony Brook University)

Mohammad Pourmatin (Role: Committee Member, Stony Brook University)

Dantu Lu (Role: External Examiner, Hong Kong University of Science and Technology; Dissertation: Innovation on Advanced Electricity Metering Infrastructure: An Empirical Analysis of Global Smart Meter Projects)

Xinyi Wu (Graduated in 2026; Role: External Examiner, University College London; Dissertation: Cost-Effective Technological Pathways for Decarbonizing the Global Iron and Steel Sector at the Plant Level)

Sumeyra Danisman (Graduated in 2026; Role: Committee Member, Stony Brook University; Dissertation: Behavioral and Policy Dynamics of Energy Efficiency Through AI-Driven Technology Adoption in SMEs)

Haozhe Yang (Graduated in 2024; Role: Committee Member, University of California, Santa Barbara; Dissertation: Socioeconomic challenges and opportunities in the low-carbon transition of the energy system)

Liqun Peng (Graduated in 2023; Role: Committee Member, Princeton University; Dissertation: Evaluating Low-carbon Technology Deployment Policies to Accelerate the Decarbonization of China's Energy System)

Mohammed Osman (Graduated in 2021; Role: Committee Chair, Stony Brook University; Dissertation: States and Carbon: A Look Ahead)

Yiyi Wang (Graduated in 2021; Role: Committee Chair, Stony Brook University; Dissertation: Encour-

aging Eco-Driving Behavior: Driver Response to Different Types of In-Vehicle Eco-Driving Feedback)

Xin Yue (Graduated in 2018; Role: Committee Chair, Stony Brook University; Dissertation: Influences of Government Championship on the Technology Innovation Process at the Project-level)

Master's Students:

Baruch College (Mariah Galindo, Kately Arriage, Nicolo Antonucci)

Graduate Center (Tiffany Hugh, Arielle Canate)

Undergraduate Students:

Macaulay Honors Student (Daniel Yadgarov); CUNY Climate Scholars (Leslieann Peers-Roman, Samid Rahman, Tenzin Sinon, Tamara Valderrama)

SERVICE

Public:

May – October, 2024, Energy Transition Sub-working Group of the U.S.-China Working Group on Enhancing Climate Action in the 2020s, Department of State

2024, China-US High-Level Dialogue on Climate and Energy Power Sector Decarbonization Working Group

2021 – 2022, New York State Climate Action Council Scoping Plan Technical Advisor Group

2019 – present, National Offshore Wind R&D Consortium's R&D Advisory Group

February 15, 2019. Testified in the New York State Senate Hearings On The Climate And Community Protection Act

Department:

2025 – present, Marx School Research Committee

2025 – present, Marx School Academic Standing Committee

2024 – 2025, Marx School Curriculum Committee

2021, Repowering the Ph.D. Program Committee

2017 – 2019, Undergraduate Curriculum Committee

2017, TMP Consortium 2017 Abstract Selection Committee

College:

2023 – 2024, Faculty Advisor, Baruch Student Climate Club

2023, Transfer Student Week Faculty Committee

2018 – 2020, College of Engineering and Applied Sciences Research Advisory Committee

2020 – 2022, College of Engineering and Applied Sciences Scholarship Committee

University:

2023, Climate Change Consortium (C3) Planning Committee

2021, Instructor, New York State Off-shore Wind Training Program

2021 – 2022, Search Committee Member, SoMAS faculty search

2021, Committee Member, Focus Group for Advanced Energy, Industrial Efficiency Technologies

2019 – 2023, Core Faculty, Tongji-Stony Brook Joint Institute for Sustainable Energy Policy

2020, Committee Member: Minor on Climate Solution

2021 – 2023, Advisory Board Member: SBU Center for Remote Internships and Experiential Learning

Community:

2017, 2018, Referee, STEM Advisor, and Judge, Clean Tech Competition 2017, 2018

2023, Faulty Advisor, NYC Climate Justice Hub

ACADEMIC SERVICE

Associate Editor:

Energy, Ecology and Environment

Editorial Board:

Resources, Conservation and Recycling

Guest Editor:

Special Issue on Clean Power Transition in China, *Resources, Conservation and Recycling*

Special Issue on Data Analytics for Energy, Water, and Environment, *IEEE-Transactions on Engineering Management*

Proposal Referee and Panel:

Sloan Foundation; National Science Foundation

Journal Referee:

Nature; Science; Proceedings of the National Academy of Sciences; Nature Climate Change; Nature Sustainability; Nature Cities; Nature Communications; Science Bulletin; Energy Policy; Environmental Science & Technology; Climate Policy; Cell Reports: Sustainability; Applied Energy; Energy Research & Social Science; Resources, Conservation and Recycling; Energy Efficiency; Energy Economics; Journal of Cleaner Production; Energy Conversion and Management, among others

Report Referee:

Renewable Energy Network (REN) 21, International Energy Agency, Center on Global Energy Policy at Columbia University, Maryland University Global Policy Center, Energy Innovation LLC, California-China Climate Institute.

Conference Referee:

The 9th biennial conference of the International Society for Industrial Ecology (ISIE) 2017

American Society of Mechanical Engineering 2014 Power Conference.

Conference session chair and/or organizer:

Association for Public Policy Analysis and Management (APPAM) 2024/2025 Fall Research Conference

American Association of Geographers Annual Meeting at Tampa 2014

Energy Management, Policy, Economics and Sustainability - energy for developing countries, International Conference on Applied Energy 2017, 2019

Seminar/Panel Organier:

2025, September 23. Climate Week NYC 2025: The Role of Nuclear Energy in Decarbonization and

Powering the AI Era. CUNY Baruch College.

2025, July 1. Macro-Energy Systems Speaker Series: Supply Chains in Energy Systems. Macro-Energy Systems.

2025, June 20. 2025 Workshop on Open Modeling Carbon Neutrality of the Power Sector. Xi'an Jiaotong University.

2024, September 24. Climate Week NYC 2024: Clean Energy Global Supply Chains Panel. CUNY Baruch College.

2024, July 19. 2024 Workshop on Open Modeling Carbon Neutrality of the Power Sector, University of California, Berkeley.

SELECTED MEDIA

30. [China Faces the Cost of Dismantling the World's Biggest Coal Sector](#), by Dan Murtaugh, *Bloomberg*, December 14, 2025.
29. [US solar tariffs linked to preventable deaths, claim researchers](#), by John Weaver, *PV Magazine*, November 5, 2025.
28. [How China Raced Ahead of the U.S. on Nuclear Power](#), by Brad Plumer and Harry Stevens, *The New York Times*, October 22, 2025.
27. [Why nuclear is now a booming industry](#), *The Economist*, September 4, 2025.
26. [In major shift, China moves toward hard carbon targets](#), by Sara Schonhardt, *E&E News*, August 5, 2024.
25. [China-US climate collaboration concerns as Xie and Kerry step down](#), by Smriti Mallapaty. *Nature*, March 12, 2024.
24. [What the U.S.-China Agreement Means for Greenhouse Gas Emissions](#), by Benjamin Storrow, *Scientific American/E&E News*, November 16, 2023.
23. [Can the U.S. and China Cooperate on Green Technology Again?](#), by Lili Pike, *Foreign Policy*, September 21, 2023.
22. [The U.S. is reducing its reliance on China for green technology. Could that slow progress on climate change?](#), by Lili Pike, *Grid News*, November 12, 2022.
21. [How Biden's made-in-America solar strategy may backfire](#), by David Iaconangelo, *E&E News*, October 27, 2022.
20. [Asia's nuclear power dilemma: Ukraine war drives energy turnarounds](#), by Dominic Faulder, *Nikkei Asia*, April 20, 2022.
19. [China creates vast research infrastructure to support ambitious climate goals](#), by Smriti Mallapaty. *Nature*, November 22, 2021.
18. [How China could be carbon neutral by mid-century](#), by Smriti Mallapaty. *Nature*, October 19, 2020.
17. [China Says It Will Stop Releasing CO2 within 40 Years](#), by Jean Chemnick, Benjamin Storrow. *Scientific American/E&E News*, September 23, 2020.
16. [Study Shows Huge Benefits of Transitioning Away from Coal in China](#), Stony Brook Press Release, August 21, 2020.
15. [Surging coal use in China threatens global CO2 goals](#), by Benjamin Storrow. *E&E News/ClimateWire*, June 9, 2020.
14. [China's Path to Clean Energy May Be Smoother than We Previously Thought](#), by Dan Gearino. *Inside*

Climate News, June 4, 2020.

13. [Study Shows Decrease in Renewable Energy Costs May Serve as an Accelerator for Clean Energy Expansion](#), Stony Brook Press Release. June 1, 2020.
12. [Bill Calls For An Emissions-Free NY By 2050](#), by Jay Shah. WSHU Public Radio Group, February 18, 2019.
11. [Global access to electricity has increased over the past two decades](#), by Michelle Bowman. Energy Information Administration, June 8, 2017.
10. [Where is the world's greenest city?](#), by Hayley Birch. *The Guardian*, April 2, 2015.
9. [Western companies gave China power projects a boost](#), by Hal Bernton. *The Seattle Times*, May 5, 2014.
8. [Chinese emission policy spells bad news for US coal exports](#), by Charles West. *China Dialogue*, October 31, 2013.
7. [China's green ministry failing in its mission](#), by Jing Li. South China Morning Post, July 9, 2013.
6. [Beijing's record smog poses health nightmare as China plans 'green' energy futures](#), by Kandy Wong. *E&E News*, February 5, 2013.
5. [Government conflicts could slow shale gas development](#), by Kandy Wong. *E&E News/ClimateWire*, May 9, 2012.
4. [Beijing Emission Cuts May Underestimate Use of Coal](#), by Kandy Wong. *Scientific American/ClimateWire*, May 7, 2012.
3. [Seeking a Pacific Northwest Gateway for U.S. Coal](#), by Stacey Schultz. *National Geographic*, October 20, 2011.
2. [As CDM Stalls, Wind Energy Drifts in China](#), by Ruidan Zhang. *Caixin News/Market Watch*, April 8, 2010.
1. [Drop in CO2 in U.S. and Power Use in China – for Now](#), by Andrew Revkin. *The New York Times/Dot Earth Blog*, May 21, 2009.

SELECTED INVITED TALKS

22. 2026, March 28. Imported Solar Photovoltaics Contributed to Health and Climate Benefits in the United States. 2026 International Conference on Natural Resources and Planning. The University of Hong Kong
21. 2026, March 14. Role of global clean energy supply chains in achieving climate goals. Brown China Summit. Brown University
20. 2026, March 4. The Cost Saving, Climate, and Health Effects of Solar PV Global Supply Chains. OMSB Seminar Series. Online: O'Malley School of Business, Manhattan University.
19. 2025, March 19. Guest Lecture: The Cost Saving, Climate, and Health Effects of Solar PV Global Supply Chains. University of California, Berkeley.
18. 2024, September 26. NYC Climate Week: Role of Clean Energy Global Supply Chains in Achieving Climate Goals. Online: Baruch College.
17. 2024, July 11. Panel: Mythbusting: Renewables and Supply Chains. Global Race to Decarbonization and Debunking Misconceptions. New York: UN Sustainable Development Solutions Network (UNSDSN), GEIDCO, Sabin Center for Climate Change Law, Climateworks, Columbia Climate School, and UNIDO.
16. 2024, September 25. Expert Panel: Subnational Climate Leadership. U.S. – China Subnational Climate

Leadership Amid Changing Dynamics. New York: California-China Climate Institute.

15. 2024, September 3. Keynote: Cost Saving, Climate, and Health Effects of Solar PV Global Supply Chains. International Conference on Applied Energy 2024. Niigata City, Japan: Keio University.
14. 2024, October 24. Power Systems Decarbonization: Technology, Policy Responses, and Implications. Online: Peking University.
13. 2024, March 27. Role of Global Clean Energy Supply Chain in Achieving Climate Goals. Seminar on China's Environment and Development. Peking University.
12. 2023, April 12. U.S. China Energy Transition. MIT Energy Conference 2023. Boston
11. 2022, November 10. Quantifying the cost savings of global solar photovoltaic supply chains. 2022 International Outstanding Young Scholars Environment Forum. Beijing: Tsinghua University.
10. 2022, March 14. Achieving power sector carbon neutrality in a low-cost renewable era. Research Seminar. Harvard University: Harvard Kennedy School Growth Lab.
9. 2022, February 17. Clean power transition in a low-cost renewable era. Guest Lecture. : Dartmouth College.
8. 2020, September 16. Accelerating renewable development in China. APERC Annual Conference 2020. Asia-Pacific Economic Cooperation.
7. 2019, April 19. Modeling the drivers of clean energy transition. Department of Engineering and Public Policy Seminar. Carnegie Mellon University.
6. 2018, October 20. China Power System Modeling Workshop: Enabling Transformation. International Forum on Energy Transitions 2018. Clean Energy Ministerial, IRENA, NREL, CNREC, and SGERI.
5. 2017, December 5. The Energy, Air Pollution, and Water Nexus Challenges in Clean Power Transition. The 3rd Sino-U.S. Energy and Environment Forum.
4. 2017, June 27. Electrification in developing countries. EIA 2017 Annual Energy Conference. Washington, DC: Energy Information Administration.
3. 2017, June 19. Electricity Access Revisited: Asian and International Experience. 40th Annual IAEE International Conference. Singapore: International Association of Energy Economy.
2. 2015, August 7. Decarbonizing China's Power Sector: Potential, Prospects, and Policy. Energy Technology Area Seminar. Lawrence Berkeley National Laboratory.
1. 2015, January 15. Decarbonizing China's Power Sector: Potential, Prospects, and Policy - the SWITCH-China model. International Electricity Modeling Workshop. U.S. Energy Information Administration.

Last updated: September 2026